simplifying the computation to

$$\mathcal{L}^{\dagger}_{2,[\![\mathrm{OYZYO}]\!]}\rho_\beta = \frac{1}{12}\left[\mathcal{L}^{\dagger}_{\mathrm{Z}},\mathcal{L}^{\dagger}_{\mathrm{Z}},\mathcal{L}^{\dagger}_{\mathrm{Y}}\right]\rho_\beta - \frac{1}{24}[\mathcal{L}^{\dagger}_{\mathrm{Y}},\mathcal{L}^{\dagger}_{\mathrm{Y}},\mathcal{L}^{\dagger}_{\mathrm{Z}}]\rho_\beta. \tag{7.29}$$

The inhomogeneity for the schemes $[\![\mathrm{OABAO}]\!]$ and $[\![\mathrm{OBABO}]\!]$ is hence a combination of the two commutator terms

$$\begin{aligned}\left[\mathcal{L}^{\dagger}_{\mathrm{A}},\mathcal{L}^{\dagger}_{\mathrm{A}},\mathcal{L}^{\dagger}_{\mathrm{B}}\right]\rho_\beta &= \left(2\beta \boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)\rho_\beta,\\ \left[\mathcal{L}^{\dagger}_{\mathrm{B}},\mathcal{L}^{\dagger}_{\mathrm{B}},\mathcal{L}^{\dagger}_{\mathrm{A}}\right]\rho_\beta &= -\left(2\beta \boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U\right)\rho_\beta,\end{aligned}$$

giving overall equations to solve

$$\begin{aligned}\mathcal{L}^{\dagger}_{\mathrm{LD}}\left(\rho_\beta f_{2,[\![\mathrm{OABAO}]\!]}\right) &= -\mathcal{L}^{\dagger}_{2,[\![\mathrm{OABAO}]\!]}\rho_\beta\\ &= \frac{\beta}{24}\left(6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)\rho_\beta,\\ \mathcal{L}^{\dagger}_{\mathrm{LD}}\left(\rho_\beta f_{2,[\![\mathrm{OBABO}]\!]}\right) &= -\mathcal{L}^{\dagger}_{2,[\![\mathrm{OBABO}]\!]}\rho_\beta\\ &= \frac{\beta}{24}\left(2\nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p} - 6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U\right)\rho_\beta.\end{aligned}$$

We can see immediately that the Fredholm alternative holds in both cases, as both equations have right-hand sides that are of odd order in p, and hence will average to 0.

If we instead consider schemes which use a central Ornstein-Uhlenbeck update, then the inhomogeneous term becomes significantly more complex, as we no longer have a significant number of cancellations in $\mathcal{L}^{\dagger}_2\rho_\beta$. Instead, we shall avoid computing the right-hand side for these schemes altogether, in favor of utilizing the following lemma [223]:

Lemma 7.1 *Consider two numerical schemes with associated operators governing the evolution of measure $\mathcal{ST}$ and $\mathcal{TS}$, with unique associated invariant distributions $\hat{\rho}_{ST}$ and $\hat{\rho}_{TS}$ respectively, such that*

$$\mathcal{ST}\hat{\rho}_{ST} = \hat{\rho}_{ST}, \qquad \mathcal{TS}\hat{\rho}_{TS} = \hat{\rho}_{TS}.$$

Under suitable ergodicity assumptions (such that in the long-time limit, all distributions converge to the invariant distribution), we have

$$\hat{\rho}_{ST} = \mathcal{S}\hat{\rho}_{TS}, \quad \textit{and} \quad \hat{\rho}_{TS} = \mathcal{T}\hat{\rho}_{ST}.$$

Proof The proof is very simple, and follows almost directly from our ergodicity assumption. If we assume that

$$\lim_{n\to\infty} (\mathcal{ST})^n \rho = \hat{\rho}_{ST},$$

for any initial distribution ρ, then we have

$$\hat{\rho}_{ST} = \lim_{n\to\infty} (\mathcal{ST})^n \hat{\rho}_{ST} = \lim_{n\to\infty} \mathcal{S} (\mathcal{TS})^{n-1} \mathcal{T} \hat{\rho}_{ST} = \mathcal{S}\hat{\rho}_{TS}.$$

A similar statement holds for $\hat{\rho}_{ST}$. □

If we return to our scheme ⟦OYZYO⟧, which has a simple inhomogeneity, then we can relate its invariant distribution to invariant distributions of schemes with a central OU update by writing its evolution operator as

$$e^{h\hat{\mathcal{L}}^\dagger_{[\![\mathrm{OYZYO}]\!]}} = \underbrace{e^{(h/2)\mathcal{L}^\dagger_{\mathrm{O}}} e^{(h/2)\mathcal{L}^\dagger_{Y}} e^{(h/2)\mathcal{L}^\dagger_{Z}}}_{\mathcal{S}} \underbrace{e^{(h/2)\mathcal{L}^\dagger_{Z}} e^{(h/2)\mathcal{L}^\dagger_{Y}} e^{(h/2)\mathcal{L}^\dagger_{\mathrm{O}}}}_{\mathcal{T}} = \mathcal{ST}.$$

By applying Lemma 7.1, we can equate the invariant distributions directly as

$$\hat{\rho}_{[\![\mathrm{ZYOYZ}]\!]} = \exp((h/2)\mathcal{L}^\dagger_Z) \exp((h/2)\mathcal{L}^\dagger_Y) \exp((h/2)\mathcal{L}^\dagger_{\mathrm{O}}) \hat{\rho}_{[\![\mathrm{OYZYO}]\!]}.$$

Rewriting our ansatz (7.27) in place of the distributions, and expanding the exponential series gives

$$\begin{aligned}
&\rho_\beta \left(1 + h^2 f_{2,[\![\mathrm{ZYOYZ}]\!]} + \ldots\right) \\
&\quad = \left(\mathrm{Id} + \frac{h}{2}\mathcal{L}^\dagger_{\mathrm{LD}} + \frac{h^2}{4}\mathcal{L}^\dagger_{1,[\![\mathrm{OYZ}]\!]} + \frac{h^2}{2}\mathcal{L}^\dagger_{\mathrm{LD}} + \ldots\right) \left(\rho_\beta \left(1 + h^2 f_{2,[\![\mathrm{OYZYO}]\!]} + \ldots\right)\right),
\end{aligned}$$

and hence equating powers we have

$$f_{2,[\![\mathrm{ZYOYZ}]\!]}\rho_\beta = f_{2,[\![\mathrm{OYZYO}]\!]}\rho_\beta + \frac{1}{4}\mathcal{L}^\dagger_{1,[\![\mathrm{OYZ}]\!]}\rho_\beta.$$

By virtue of (7.20), we have

$$\mathcal{L}^\dagger_{1,[\![\mathrm{OAB}]\!]}\rho_\beta = \frac{1}{2}\mathcal{L}^\dagger_H (g\rho_\beta) \qquad \mathcal{L}^\dagger_{1,[\![\mathrm{OBA}]\!]}\rho_\beta = -\frac{1}{2}\mathcal{L}^\dagger_H (g\rho_\beta),$$

where recall $g(\boldsymbol{q},\boldsymbol{p})\rho_\beta = \mathcal{L}^\dagger_{\mathrm{A}}\rho_\beta = \beta \boldsymbol{p}^T \boldsymbol{M}^{-1} \nabla U(\boldsymbol{q})\rho_\beta$. Computing

$$\mathcal{L}^\dagger_H (g\rho_\beta) = \beta \left(|\nabla U(\boldsymbol{q})|^2 - \boldsymbol{p}^T \nabla^2 U(\boldsymbol{q}) \boldsymbol{p}\right) \rho_\beta,$$